Name: ______________________________ Date: _________________ Class: _______________

MyDesign Portfolio

Element G Initial Template

Element G: Construction of a Testable Prototype

Final Prototype Iteration Explanation *(G1)*

Date: 2/29/24

Photos with captions of your full construction process from day 1 of building through to the last day of constructing your final prototype iteration:


Details about all steps of construction (add step numbers as needed), including the tools and manufacturing methods used:

	Details	Tools and Manufacturing Methods used
Step 1	Cut the cardboards in one feet each	Utility knife, cutting vertical lines
Step 2	Fold the cardboards, making it round	Did it with our hands, folding the sides from left to right
Step 3	Put some tape when you fold it	Cut the tape with our fingers in medium pieces
Step 4	Then use the tape to put the pieces together, one on top of another	Put the pieces together and the tape around

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Step 5	Make a grabber to put it on top	Cut it to look like a claw
Step 6	Tape the grabber on top of the prototype	Using a utility knife, cut the cardboard in a rectangle and
Step 7	Bend a small piece of cardboard and put it inside of another folded one, to make it thicker	Bend it with our hands, from left to right.

Material name	**Description**	**Where material was purchased**	**Cost per unit**	**# units**	**Total cost per item**
Cardboard	Recycled cardboard from boxes	In class	No charge, classroom supplies	3 big cardboards	No charge, classroom supplies
Tape	Narrow strip material	In class	No charge, classroom supplies	5 rolls of tape	No charge, classroom supplies
Velcro	Half a velcro	In class	No charge, classroom supplies	5 cm	No charge, classroom supplies
Wood	3 ft long pieces of wood	In Class	No charge, classroom supplies	3 pieces	No charge, classroom supplies

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Detailed sketches, which might include CAD if you have learned it in your class:

Construction of the Final Prototype *(G2)*

Design Requirement	How will this prototype allow you to collect objective, measurable data?
The device must be able to reach the badminton birdies on the window and have enough height to do so.	We will first test in our class, putting the birdies in tall places and testing if our prototype can reach it, and after we succeed, we could test in the gym.
The device must be able to endure getting hit by sports equipment without sustaining damage and not break in half in testing.	We consider that maybe some birdies can fall and hit our prototype, so we will test if he is resistant enough to not break if this happens.
Our device must be able to be transported into a gym without problems.	We will transport the prototype to the gym and see how easily he can be transported. Considering weight, height, and sizes.

Testing or Modeling of Attributes *(G3, G4)* and Addressing Non-Testable Attributes *(G4)*

Attribute	**Description**	**Describe your prototype's functionality to test this attribute**
Height	The prototype is tall enough to remove birdies	We went to the stairs window and successfully tested

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	from windows	
Durability	The prototype is resistant enough	We made the prototype very thick, reinforcing the cardboard and using wood sticks to make it more resistant
Mobility/Transportation	The prototype is easy to transport	The prototype in dismountable, what makes it easy to transport between places

Addressing Non-Testable Attributes *(G4)*

Attribute	**Justification for why this attribute cannot be modeled or tested with your prototype**
	There are no non-testable attributes to this module

Element G: Construction of a Testable Prototype						
	5	4	3	2	1	0
1. Final Prototype Iteration Explanation. The Final Prototype Iteration is ______________.	clearly and *fully* explained	*clearly and adequately explained*	*adequately explained*	*somewhat explained but with gaps*	*only minimally* explained	is *unclear* or is *missing* altogether

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2. Construction of the Final Prototype. The Final Prototype Iteration is _____________.	constructed with enough detail to assure that objective data on *all* or nearly all design requirements could be determined	constructed with enough detail to assure that objective data on *many or nearly all* design requirements could be determined	constructed with enough detail to assure that objective data on *some* design requirements could be determined	constructed with enough detail to assure that objective data on at least a *few* design requirements could be determined	*not* constructed with enough detail to assure that objective data on at least *one* design requirement could be determined	*not applicable* to any design requirements and/or functional claims
3.Testing or Modeling of Attributes. _____________ attributes (sub-systems) of the unique solution that can be tested or modeled mathematically are addressed.	*All*	*Most*	*Some*	*A Few*	*No more than one*	*None of the*
4.Addressing non-Testable Attributes. _____________ justification is provided for those that cannot be tested or modeled mathematically and thus require expert review.	A *well* supported	A *generally* supported	An *adequately* supported	*Developing*	*Insufficient*	*No*